



2020 ASSESSMENT TASK 1 DECEMBER

Mathematics Extension 1

**General
Instructions**

- Reading time – 5 minutes
- Working time – 1 hour
- Write using black pen
- Calculators approved by NESA may be used
- In Questions 1 – 4, show relevant mathematical reasoning and/or calculations

**Total marks:
38**

Attempt all Questions 1 – 4 (pages 2-6)

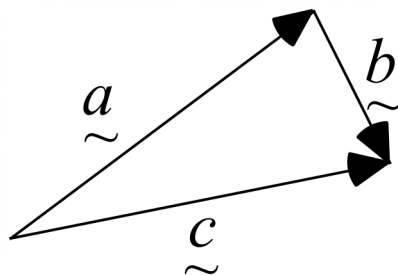
Question 1 Start on the appropriate page of your answer booklet

10 marks

a) Simplify $\vec{AB} + \vec{BC} - \vec{DC}$

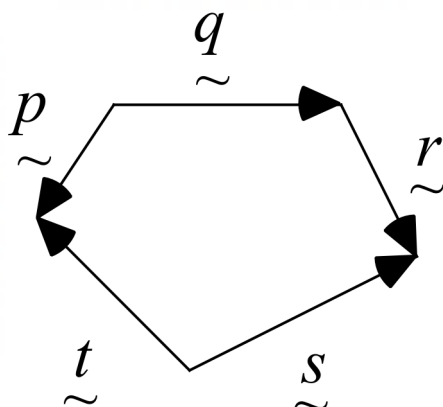
2

b) A vector equation for the diagram below is $\vec{a} + \vec{b} = \vec{c}$.



Write a possible vector equation for the following diagram.

1



c) If $a = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ and $b = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$, find

(i) $\vec{a} \cdot \vec{b}$

1

(ii) $\vec{a} - 2\vec{b}$

1

(iii) $\hat{a}$

2

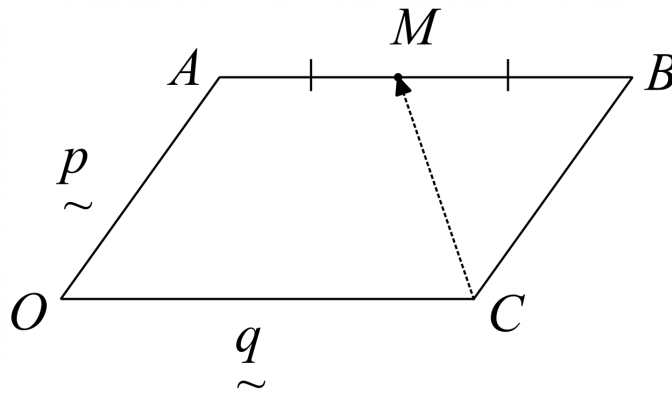
d) Prove, by mathematical induction, that $3^{2n-1} + 2^{n+1}$ is divisible by 7 for all positive integers, $n \geq 1$. **3**

Question 2 Start on the appropriate page of your answer booklet

8 marks

a) Find the vector projection of $\underline{a} = \underline{i} + 3\underline{j}$ onto $\underline{b} = 3\underline{i} - 4\underline{j}$. 2

b) $OABC$ is a parallelogram. Let $\underline{p} = \overrightarrow{OA}$ and $\underline{q} = \overrightarrow{OC}$. Let M be the midpoint of AB .



(i) Express $\overrightarrow{OB}$ and $\overrightarrow{OM}$ in terms of $\underline{p}$ and $\underline{q}$. 2

(ii) If X is a point on CM and $\overrightarrow{CX} = \frac{2}{3} \overrightarrow{CM}$, express $\overrightarrow{CX}$ in terms of $\underline{p}$ and $\underline{q}$. 1

(iii) Find $\overrightarrow{OX}$ and hence show that X lies on the line segment OB . 2

(iv) Find the ratio in which the median CM divides the diagonal OB . 1

Question 3 Start on the appropriate page of your answer booklet**10 marks**

a) If $|\underline{a}| = 5$, $|\underline{b}| = 6$ and the angle between $\underline{a}$ and $\underline{b}$ is 45° , find $\underline{a} \cdot \underline{b}$ **1**

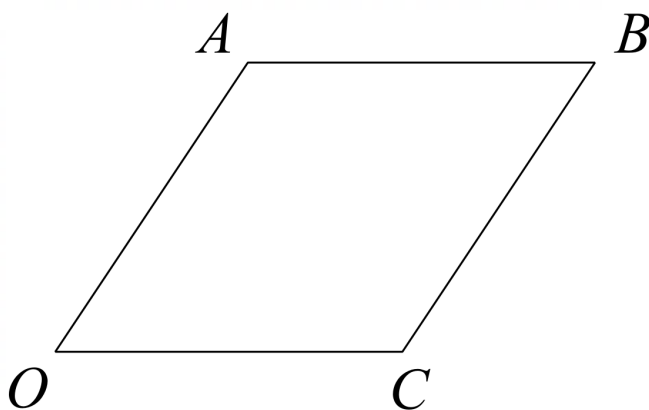
b) Let $S_n = \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!}$, for all positive integer n .

(i) Prove, using mathematical induction, that $S_n = 1 - \frac{1}{(n+1)!}$ for all positive integers $n \geq 1$. **3**

(ii) Find the value of $\lim_{n \rightarrow \infty} S_n$. **1**

(iii) Find the smallest positive integer n such that $|S_n - 1| < 10^{-6}$. **1**

c) Consider a rhombus $OABC$ where $\overrightarrow{OA}$ is defined as $\underline{m} + \underline{n}$ and $\overrightarrow{OC}$ is defined as $\underline{n}$.



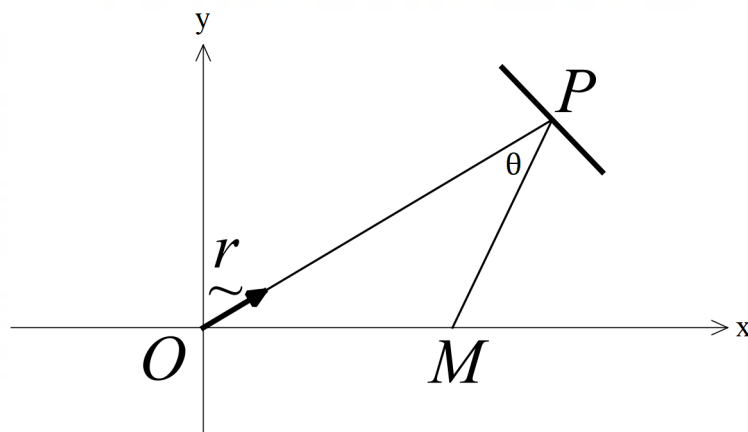
Using these vector definitions, prove that the diagonals of $OABC$ are perpendicular. **4**

Question 4 Start on the appropriate page of your answer booklet

10 marks

- a) The diagram shows the path of a light beam from its source at O in the direction

of the vector $\underline{r} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$.



At point P , the beam is reflected by an adjustable mirror and meets the x -axis at M . The position of M varies, depending on the adjustment of the mirror at P .

- (i) Given that $\overrightarrow{OP} = 4\underline{r}$ find the coordinates of P . **1**

- (ii) The point M has coordinates $(k,0)$. Find an expression, in terms of k , for vector $\overrightarrow{PM}$. **1**

- (iii) Find the magnitude of the vectors $\overrightarrow{OP}$ and $\overrightarrow{PM}$. **2**

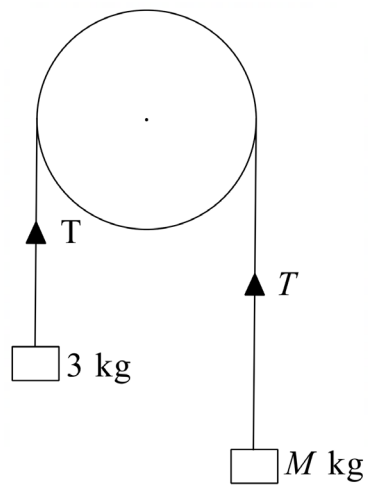
- (iv) Hence, using vector techniques and the information in parts (i) to (iii), find the value of

- (1) k if $\theta = 90^\circ$. **1**

- (2) θ if M has coordinates $(9,0)$. Answer to the nearest degree. **2**

Question 4 continues on the next page

- b) Two blocks of masses 3 kg and M kg (where $M > 3$) are connected by a light inextensible string passing over a smooth fixed pulley. The two masses are at rest before being released.



Given that the tension in the string is 37.5 N, calculate the value of M .

3

(Use $g = 9.8 \text{ ms}^{-2}$)

End of paper

EXTENSION 1 DECEMBER ASSESSMENT

1a) $\vec{AB} + \vec{BC} - \vec{DC}$
 $= \vec{AC} + \vec{CB}$
 $= \vec{AB}$

- ② correct solution
- ① simplifies two of the vectors or adds $\vec{CB}$

b) $\underline{q} + \underline{r} - \underline{s} + \underline{t} = \underline{p}$ (or equivalent)

- ① correct answer

c) i) $\begin{pmatrix} 3 \\ -2 \end{pmatrix} - \begin{pmatrix} -1 \\ 5 \end{pmatrix} = -3 - 10$
 $= -13$

- ① correct answer

ii) $\underline{a} - 2\underline{b} = \begin{pmatrix} 3 \\ -2 \end{pmatrix} - 2 \begin{pmatrix} -1 \\ 5 \end{pmatrix}$
 $= \begin{pmatrix} 3+2 \\ -2-10 \end{pmatrix}$
 $= \begin{pmatrix} 5 \\ -12 \end{pmatrix}$

- ① correct answer

iii) $\hat{a} = \frac{1}{\sqrt{3^2 + (-2)^2}} \begin{pmatrix} 3 \\ -2 \end{pmatrix}$
 $= \frac{1}{\sqrt{13}} \begin{pmatrix} 3 \\ -2 \end{pmatrix}$

- ② correct solution
- ① finds magnitude of $\underline{a}$

d) Test $n=1$
 $3^{2-1} + 2^{n-1}$
 $= 3 + 4$
 $= 7 \times 1$

$\therefore$ True for $n=1$

Assume true for $n=k$ (k is a positive integer)

$3^{2k-1} + 2^{k+1} = 7P$ (where P is an integer)

For $n=k+1$ we wish to prove

$3^{2(k+1)-1} + 2^{(k+1)+1} = 7Q$ (where Q is an integer)

ie $3^{2k+1} + 2^{k+2} = 7Q$

$3^{2k+1} + 2^{k+2} = 3^2 3^{2k-1} + 2^{k+2}$

$= 9(7P - 2^{k+1}) + 2 \times 2^{k+1}$ (using assumption)

- ③ correct proof
- ② Successfully does 3 of 4 key parts below
- ① Successfully does 2 of 4 key parts below
- 4 key parts
- ① proves true for $n=1$
- ② states assumption and what must be proven
- ③ uses assumption
- ④ correctly proves required statement

$$= 7 \times 9P - 9 \times 2^{k+1} + 2 \times 2^{k+1}$$

$$= 7 \times 9P - 7 \times 2^{k+1}$$

$$= 7(9P - 2^{k+1})$$

$$= 7Q \text{ where } Q = 9P - 2^{k+1}, \text{ which is an integer as } P, k \text{ are integers}$$

$\therefore$ If true for $n=k$, it is true for $n=k+1$. But it is true for $n=1$, therefore true for $n=1+1=2, 3, 4, \dots$ and so on for all positive integers $n \geq 1$

Q2 a) $\text{proj}_{\underline{b}} \underline{a} = \frac{\underline{a} \cdot \underline{b}}{\underline{b} \cdot \underline{b}} \underline{b}$

$$= \frac{3 - 12}{25} (3\underline{i} - 4\underline{j})$$

$$\begin{cases} = -\frac{9}{25} (3\underline{i} - 4\underline{j}) \\ = -\frac{27}{25} \underline{i} + \frac{36}{25} \underline{j} \end{cases}$$

② correct solution

① progress towards

2 b) i) $\vec{OB} = \vec{OA} + \vec{AB}$

$$= \vec{OA} + \vec{OC} \text{ (opposite sides of a parallelogram)}$$

$$= \underline{p} + \underline{q}$$

② correct solution

① correct expression for one of the vectors

$$\vec{OM} = \vec{OA} + \vec{AM}$$

$$= \underline{p} + \frac{\underline{q}}{2}$$

ii) $\vec{OC} + \vec{CM} = \vec{OM}$

$$\vec{CM} = \underline{p} + \frac{1}{2} \underline{q} - \underline{q}$$

$$= \underline{p} - \frac{1}{2} \underline{q}$$

$$\begin{aligned} \vec{CX} &= \frac{2}{3} \left(\underline{p} - \frac{1}{2} \underline{q} \right) \\ &= \frac{1}{3} (2\underline{p} - \underline{q}) \end{aligned} \quad \left. \vphantom{\begin{aligned} \vec{CX} &= \frac{2}{3} \left(\underline{p} - \frac{1}{2} \underline{q} \right) \\ &= \frac{1}{3} (2\underline{p} - \underline{q}) \end{aligned}} \right\}$$

① correct answer

2b iii) $\vec{OX} = \vec{OC} + \vec{CX}$
 $\vec{OX} = \frac{2}{3} \vec{p} - \frac{1}{3} \vec{q} + \vec{q}$
 $= \frac{2}{3} \vec{p} + \frac{2}{3} \vec{q}$
 $= \frac{2}{3} (\vec{p} + \vec{q})$
 $= \frac{2}{3} \vec{OB}$

$\therefore$ Since $\vec{OX} = \frac{2}{3} \vec{OB}$, $\vec{OX} \parallel \vec{OB}$ and lies on $\vec{OB}$ as it shares a common point O.

② Correct solution
 ① finds $\vec{OX}$ in terms of $\vec{p}$ and $\vec{q}$

iv) X divides OB in the ratio 2:1 ① correct answer

3a $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$
 $= 5 \times 6 \times \cos 45^\circ$
 $= \frac{30}{\sqrt{2}}$
 $= 15\sqrt{2}$

① correct solution

3b i) Test $n=1$
 $S_1 = \frac{1}{(1+1)!}$
 $= \frac{1}{2}$
 RHS = $1 - \frac{1}{(1+1)!}$
 $= 1 - \frac{1}{2}$
 $= \frac{1}{2}$
 $= \text{LHS}$

$\therefore$ True for $n=1$

Assume true for $n=k$

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} = 1 - \frac{1}{(k+1)!}$$

For $n=k+1$ we wish to prove

$$\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!} = 1 - \frac{1}{(k+2)!}$$

③ Correct proof
 ② Successfully does 3 of 4 key parts below
 ① Successfully does 2 of 4 key parts below
 (I) Proves true for $n=1$
 (II) States assumption and what must be proven
 (III) Uses assumption
 (IV) Proves required result

3b (continued)

$$\text{Now } \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{k}{(k+1)!} + \frac{k+1}{(k+2)!}$$

$$= 1 - \frac{1}{(k+1)!} + \frac{k+1}{(k+2)!} \quad (\text{using assumption})$$

$$= 1 + \left(\frac{k+1}{(k+2)!} - \frac{k+2}{(k+2)(k+1)!} \right)$$

$$= 1 + \frac{k+1-k-2}{(k+2)!}$$

$$= 1 - \frac{1}{(k+2)!} \text{ as required.}$$

Therefore, if true for $n=k$, it is true for $n=k+1$. Since true for $n=1$, by mathematical induction, it is true for $n=2, 3, 4$ and so on for all positive integers.

3b ii) $\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{(n+1)!} \right)$

① Correct solution

$$= 1 - 0$$

$$= 1$$

3b iii) $|S_n - 1| < 10^{-6}$

① Correct solution

$$\left| \frac{-1}{(n+1)!} \right| < \frac{1}{10^6}$$

$$(n+1)! > 10^6$$

$$9! = 362880$$

$$10! = 3628800$$

$$> 10^6$$

$$(9+1)! > 10^6$$

$\therefore n=9$ is the smallest positive integer

$$3 \quad c) \quad \vec{CA} = \vec{OA} - \vec{OC}$$

$$= \underline{a} + \underline{b} - \underline{b}$$

$$= \underline{a}$$

$$\vec{OB} \cdot \vec{CA} = (\underline{a} + \underline{b} + \underline{b}) \cdot \underline{a}$$

$$= (\underline{a} + 2\underline{b}) \cdot \underline{a}$$

$$= \underline{a} \cdot \underline{a} + 2\underline{a} \cdot \underline{b} \quad *$$

Since OABC is a rhombus

$$|\vec{OA}| = |\vec{OC}|$$

$$|\underline{a} + \underline{b}| = |\underline{b}|$$

$$(\underline{a} + \underline{b}) \cdot (\underline{a} + \underline{b}) = \underline{b} \cdot \underline{b}$$

$$\underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{b} + \underline{b} \cdot \underline{a} + \underline{b} \cdot \underline{b} = \underline{b} \cdot \underline{b}$$

$$\underline{a} \cdot \underline{a} + 2\underline{a} \cdot \underline{b} = 0$$

$$\text{From } * \quad \vec{OB} \cdot \vec{CA} = 0$$

$\therefore OB \perp CA$, so the diagonals of OABC are perpendicular

$$4a \quad i) \quad \vec{OP} = 4 \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 12 \\ 4 \end{pmatrix}$$

$$\therefore P = (12, 4)$$

① Correct answer

$$ii) \quad \vec{PM} = \vec{OM} - \vec{OP}$$

$$= \begin{pmatrix} k \\ 0 \end{pmatrix} - \begin{pmatrix} 12 \\ 4 \end{pmatrix}$$

$$\vec{PM} = \begin{pmatrix} k-12 \\ -4 \end{pmatrix}$$

① Correct answer

$$iii) \quad |\vec{OP}| = \sqrt{12^2 + 4^2}$$

$$= \sqrt{160}$$

$$= 4\sqrt{10}$$

$$|\vec{PM}| = \sqrt{(k-12)^2 + (-4)^2}$$

$$= \sqrt{k^2 - 24k + 160}$$

② Correct solution

① Finds one correct magnitude

4a iv) 1. If $\theta = 90^\circ$ $\vec{PM} \cdot \vec{OP} = 0$

$$\begin{pmatrix} k-12 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 12 \\ 4 \end{pmatrix} = 0$$

$$12(k-12) - 4 \times 4 = 0$$

$$12k = 160$$

$$\therefore k = \frac{40}{3}$$

① Correct answer

2. If $k=9$

$$\cos(180^\circ - \theta) = \frac{\begin{pmatrix} -3 \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 12 \\ 4 \end{pmatrix}}{\sqrt{160} \times \sqrt{25}}$$

$$= \frac{-52}{20\sqrt{10}}$$

$$= -0.82219\dots$$

$$\theta = 145.30\dots$$

$$\theta = 35^\circ \text{ (nearest degree)}$$

② Correct solution

① finds obtuse θ or significant progress

4 b) For the 3kg block

$$F = ma$$

$$3a = T - 3g$$

$$3a = 37.5 - 3g$$

$$a = 12.5 - 9.8$$

$$a = 2.7 \text{ ms}^{-2}$$

③ Correct solution

② Finds valid equations for both blocks and resolves into a value for m .

① Finds a valid equation for the 3kg block and obtains 8.1 N or 2.7 ms^{-2} .

For the M kg block

$$F = ma$$

$$Ma = Mg - T$$

$$M(9.8 - 2.7) = 37.5$$

$$M = 5.28 \text{ kg (2 decimal places)}$$